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Reinforcement Learning
Assignment 1

Q1 In a state s, a policy π chooses action 'a_1' with probabili 0.6 and action a_2 with 0.4.

If a_1 is chosen, reward is 4 and moves to state S_1 where $V^{\pi}(S_1) = 5$
If a_2 is chosen, reward is 1 and moves to state S_2 where $V^{\pi}(S_2) = 8$.

Discount factor $\gamma = 0.9$. Compute $V^{\pi}(s)$.

Ans 1 Using Bellman expectition equation:

$$V^{\pi}(s) = \sum_{a \in A} \pi(a|s) \sum_{s',r} (p(s', r|s, a) [r + \gamma V^{\pi}(s')$$

Since transition outcomes for each action are deterministic here, this simplifies to expected value across 2 possible actions:

$$V^\pi(s) = \pi(a_1|s)\,[r_1 + \gamma V^\pi(s_1)] + \pi(a_2|s)\,[r_2 + \gamma V^\pi(s_2)]$$

Given:

$\pi(a_1|s) = 0.6 \qquad \pi(a_2|s) = 0.4$

For action a_1: $r_1 = 4$, next state value $V^\pi(s_1) = 5$

For a_2: $r_2 = -1$, $V^\pi(s_2) = 8$

Discount factor $\gamma = 0.9$

$$\therefore V^\pi(s) = 0.6\,(4 + 0.9 \times 5) + 0.4\,(-1 + 0.9 \times 8)$$

$$= \underline{7.58}$$

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Q2 & 3: From state s, agent has two outcomes: action a_1 having immediate reward 5, next state ~~$V^*(s') = 8$~~ terminal.

Or a_2: immediate reward 1, next state $V^*(s) = 8$.

Discount factor $\gamma = 0.75$.

Which action is optimal according to Bellman optimality principle?

Ans $Q^*(s,a) = R(s,a) + \gamma V^*(s')$

For action a_1:

$Q^*(s, a_1) = 5 + 0.75 \times 0 = 5$

For a_2:

$Q^*(s, a_2) = 1 + 0.75 \times 8 = 7$

So state s has better action (higher optimal value). a_2

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Q4 Consider 2-state Markov (MDP) with state S_1, S_2. A fixed deterministic policy is followed.

$S_1 \longrightarrow S_2$ with reward 3
$S_2 \longrightarrow S_1$ with reward 1

Discount factor is $\gamma = 0.8$.
What is $V^{\pi}(S_1)$ by solving Bellman equations?

Ans $V^{\pi}(s) = R(s, \pi(s)) + \gamma \sum_{s'} P(s' | s, \pi(s)) V^{\pi}(s')$

$V^{\pi}(s_1) = 3 + 0.8\, V^{\pi}(s_2)$
$V^{\pi}(s_2) = 1 + 0.8\, V^{\pi}(s_1)$

This is linear system of equations.
Substituting:

$V^{\pi}(s_1) = 3 + 0.8(1 + 0.8\, V^{\pi}(s_1))$
$\Rightarrow (1 - 0.64) V^{\pi}(s_1) = 3 + 0.8$
$\Rightarrow V^{\pi}(s_1) = 3.8/0.36 = \underline{10.56}$

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Q6 From state s, action a has stochastic outcomes:

$P(s_1|s, a) = 0.7$, $R(s, a, s_1) = 2$
$P(s_2|s, a) = 0.3$, $R(s, a, s_2) = -2$

Given $V^\pi(s_1) = 6$, $V^\pi(s_2) = 4$, $\gamma = 0.5$
Compute $Q^\pi(s, a)$

Ans $Q^\pi(s, a) = 0.7(2 + 0.5 \times 6)$
$+ 0.3(-2 + 0.5 \times 2)$
$= 3.5$

Q7 Consider following episodes observed in undiscounted episodic task:

Episode 1: $A \xrightarrow{2} B \xrightarrow{-1} A \xrightarrow{4} T$
" 2: $B \xrightarrow{1} A \xrightarrow{3} T$
3: $A \xrightarrow{0} T$

Using returns from episodes, what are First Visit and Every Visit MC estimates of $V(A)$?

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Ans Returns of ~~each~~ each instance of A (since no discount):

→ Episode 1:
* First Visit $= 2-1+4 = 5$
* Second Visit $= 4$

→ Episode 2: 3

→ Episode 3: 0

Estimates of $V(A)$:

1) First Visit MC: $\frac{5+3+0}{3} = 2.67$

2) Every Visit MC: $\frac{5+4+3+0}{4} = 3$

Q9 Returns observed for state X over first 3 visits are:

4, 7, 1

Starting with $V_0(X)=0$, use ~~the~~ incremental sample-average MC updates with $\alpha_n = 1/n$

What is final ~~update~~ estimate after updates?

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Ans For each update: ~~$V^{\pi}(s)$~~ =

$$V_{n+1}(s) = V_n(s) + \alpha_n(G - V_n(s)$$

Updates:

→ First: $0 + \frac{1}{1}(4-0) = 4$

→ Second: $4 + \frac{1}{2}(7-4) = 5.5$

→ Third: $5.5 + \frac{1}{3}(1-5.5) = 3$

∴ Final estimate is 3.

Q11 For state: s there are actions a, ~~b~~. For action a:

$P(S_1 | S, a) = 0.5$, $R(S, a, S_1) = 2$
$P(S_2 | S, a) = 0.5$, $R(S, a, S_2) = 0$

For action b:

$P(S_1 | S, b) = 1$, $R(S, b, S_1) = 1$

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Given: $V^*(s_1) = 10$, $V^*(S_2) = 4$, $\gamma = 0.8$

Compute $V^*(s)$ using Bellman optimality equations.

Ans $$V^*(s) = \max_a \sum_{s'} p(s'|s, a)\left[R(s, a, s') + \gamma V^*(s')\right]$$

$$\cancel{\Rightarrow V^*(s) = \max_a \{(0.5(2+0.8\times 10) + 0.5(0+0.8\times 4), (0.5(10) + 0.5(32)\}}$$

$$\cancel{= \max_a}$$

Action value for a:

$$Q^*(s, a) = 0.5(2+0.8\times 10) + 0.5(0+0.8\times 4) = 6.6$$

Action value for b:

$$Q^*(s, b) = 1(1+0.8\times 10) = 9$$

Maximum action value : 9

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Q13 $\gamma = 0.9$.

Shift 1: $N \xrightarrow{C,3} W \xrightarrow{I,1} N \xrightarrow{C,3} W \xrightarrow{R,-2} T$

Shift 2: $W \xrightarrow{C,2} N \xrightarrow{I,1} W \xrightarrow{C,4} T$

Shift 3: $N \xrightarrow{C,4} W \xrightarrow{I,1} N \xrightarrow{C,4} T$

Every Visit State Value Estimate
$V(W) = ?$

Ans List of returns of W
for each instance:

→ Shift 1:
* First Visit: $1 + 0.9(3) + 0.9^2(-2)$

$= 2.08$

* Second: -2

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→ Shift 2:

* First: $2 + 0.9(1) + 0.9^2(4)$
  $= 6.14$
* Second: 4

→ Shift 3 Visit: $1 + 0.9(4) = 1.36$

$\therefore$ Estimate $= \frac{2.08 \; \cancel{-2} + 6.14 + 4 + 1.36}{5}$

$= \underline{2.316}$

Q14 From Q13,
now $Q_0(W, I) = 3.00$ [initial action value estimate]

With fixed $\alpha = 0.5$,
Incrementally update $Q(W, I)$
(chronological order)
by processing all (W, I)
occurances.
Final estimate of $Q(W, I)$?

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Ans $Q_0(W, I) = 3$ (initial)

$Q_1(W, I) = 3 + 0.5(1 - 3)$

$= 2$ (update)

$Q_2(W, I) = 2 + 0.5(1 - 2) = 1.5$ (update)

Final answer: 1.5

Here first occurance was in shift 1, next in shift 3.